



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

PRE - SCORE

11-11-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

TEST DATE : 11-11-2021

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	B	B	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,C	A	A,B,D	A,B,D	A,B	A,B,D	B,C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.20	7.00	5.00	105.00	8.00	9.00		

PART-2 : CHEMISTRY

SECTION-I (I)	Q.	1	2	3	4				
	A.	A	C	A	C				
SECTION-I (II)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,D	B,C	A,B,C	A,B,C,D	A,C,D	D	C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	455.00	1.00	8.00	3.00	3.00	5.00		

PART-3 : MATHEMATICS

SECTION-I (I)	Q.	1	2	3	4				
	A.	A	B	C	A				
SECTION-I (II)	Q.	5	6	7	8	9	10	11	12
	A.	A,C,D	A,D	A,B,C	A,B,C	B,D	A,B,C	A,C	A,B,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	0.60	0.25	8.68	0.75	0.50	0.83		

HINT – SHEET

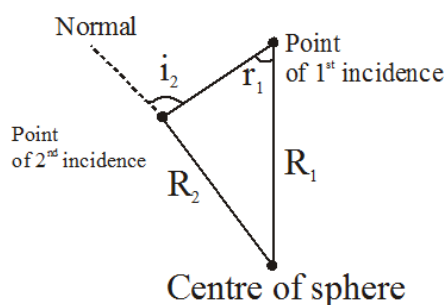
PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (B)**

$$\mu_1 \sin i_1 = \mu_2 \sin r_1$$

$$\text{Also, } \frac{\sin i_2}{R_1} = \frac{\sin r_1}{R_2}$$



$\Rightarrow$ Eliminating $\sin r_1$, we get ;

$$\mu_1 R_1 \sin i_1 = \mu_2 R_2 \sin i_2$$

2. **Ans (B)**

$$\begin{aligned} \text{Shift } x &= t \left(1 - \frac{1}{\mu} \right) \\ &= 6 \left(1 - \frac{1}{\frac{3}{2}} \right) = 2 \text{ cm} \end{aligned}$$

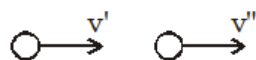
$$\therefore u = 60 \text{ and } v = 30 \text{ cm}$$

$\therefore$ final answer (B)

3. **Ans (B)**

$$\frac{(\mu - 1)t}{\lambda} = 7$$

4. Ans (D)



λ same $\Rightarrow p$ same

$$mv - \frac{2mv}{2} = mv' + 2mv''$$

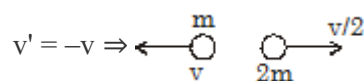
$$1 = \frac{v'' - v'}{v + \frac{v}{2}}$$

$$v' = -2v''$$

$$v'' - v' = \frac{3v}{2}$$

$$3v'' = \frac{3v}{2}$$

$$v'' = \frac{v}{2}$$



PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,C)

From force equation

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{Ze \cdot e}{r^2} \dots(i)$$

From Bohr's assumption,

$$mvr = \frac{nh}{2\pi} \dots(ii)$$

Solving equation (i) & (ii)

$$r \propto \frac{n^2}{Z} \text{ \& \; } v \propto \frac{Z}{n}$$

$$E \propto \frac{Z^2}{n^2}$$

6. Ans (A)

$$\lambda_{\min} = \frac{1500}{1 - 1/p^2} = 1500 \text{ \AA} \Rightarrow \text{most energetic}$$

$$\lambda_{\max} = \frac{1500 \times 4}{3} = 2000 \text{ \AA} \Rightarrow \text{least energetic}$$

$$\frac{\lambda_{\max}}{\lambda_{\min}} = \frac{4}{3}$$

For energy in ground state, transition occurs from ∞ to $n = 1$

$$\therefore \lambda_{\min} = 1500$$

$$\therefore E_{\text{ground state}} = \frac{hc}{\lambda_{\min}} = 8.27 \text{ eV}$$

1st line of Balmer series will corresponds to transition from $n = 3$ to $n = 2$.

$$\text{Energy from transition from 3 to 1} = \frac{hc}{\left(\frac{1500 \times 9}{8}\right)}$$

$$\text{Energy from transition from 2 to 1} = \frac{hc}{\left(\frac{1500 \times 4}{3}\right)}$$

$\therefore$ 1st line of Balmer series will have energy

$$\text{given by } hc \left[\frac{8}{1500 \times 9} - \frac{3}{1500 \times 4} \right]$$

7. Ans (A,B,D)

We have half life of the decay is 3.8 day.

$\therefore$ Amount left after 11.4 days = 3 half life

$$= \frac{N_0}{2^3} = \frac{N_0}{8}$$

Activity of radon after 7.6 days = 2 half

$$\text{life} = \frac{A_0}{4}.$$

Rules of radioactive decay applies for large number of sample product.

Po will be more stable

8. Ans (A,B,D)

Since piston is movable

$\Rightarrow$ force on piston in equilibrium must be zero.

$\therefore$ Pressure in both chamber must be same

By conserving number of moles

$$\frac{PV}{T} = \frac{P_f V_f}{T} \dots(i)$$

$$\& \frac{2P \times 2V}{T} = \frac{P_f (3V - V_f)}{T} \dots(ii)$$

Solving (i) & (ii), $V_f = \frac{3V}{5}$ (volume of chamber-I)

Volume of chamber-II will be $3V - V_f = \frac{12V}{5}$

$$\text{Also } P_f = \frac{5}{3}P$$

9. **Ans (A,B)**

Since surface area is same

⇒ They radiate at same rate initially and also absorb at same rate initially

Since mass of two sphere are different

⇒ temperature at any other instant will be different

11. **Ans (B,C)**

$$\text{At A : } \Delta X = \frac{10 \times 10^{-3} \times 5 \times 10^{-3}}{2} - \frac{10^{-3} \times 10 \times 10^{-3}}{1}$$

$$= 15 \times 10^{-6} \text{ m} = 30\lambda$$

So maxima at A

$$\text{At B : } \Delta X = 35 \times 10^{-6} \text{ m} = 70\lambda$$

maxima at B

$$\text{At O : } \Delta X = 10 \times 10^{-6} = 20\lambda$$

maxima at O

For central maxima at 'O'

$$(\mu - 1)t = 10^{-5}$$

$$t = 2 \times 10^{-5} \text{ m}$$

12. **Ans (A,B,C)**

$$C_P - C_V = \left(\frac{dQ}{dT} \right)_P - \left(\frac{dU}{dT} \right)_V = P \frac{dV}{dT}$$

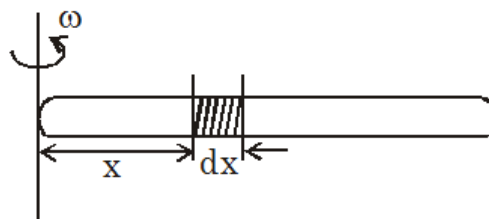
$$\alpha = \frac{1}{3V} \frac{dV}{dT} \Rightarrow \frac{dV}{dT} = 3\alpha V = 3\alpha \frac{M}{\rho}$$

$$\therefore C_P - C_V = 3\alpha \frac{M}{\rho} P$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (2.20)**



$$\text{Elongation in } dx \text{ length} = \frac{m\omega^2(\ell^2 - x^2)dx}{2\ell AY}$$

∴ Elongation in 1st half

$$= \int_0^{\ell/2} \frac{m\omega^2}{2AY\ell} (\ell^2 - x^2) dx$$

$$= \frac{m\omega^2}{2AY\ell} \left[\ell^2 \times \frac{\ell}{2} - \frac{\ell^3}{3 \times 8} \right] = \frac{m\omega^2}{2AY\ell} \left[\frac{11\ell^3}{8 \times 3} \right]$$

$$\text{Elongation in 2nd half} = \int_{\ell/2}^{\ell} \frac{m\omega^2}{2AY\ell} (\ell^2 - x^2) dx$$

$$= \frac{m\omega^2}{2AY\ell} \left[\ell^2 x - \frac{x^3}{3} \right]_{\ell/2}^{\ell}$$

$$= \frac{m\omega^2}{2AY\ell} \left[\left\{ \ell^2 - \frac{\ell^3}{3} \right\} - \left\{ \frac{\ell^3}{2} - \frac{\ell^3}{24} \right\} \right]$$

$$= \frac{m\omega^2}{2AY\ell} \left[\frac{2}{3}\ell^3 - \frac{11}{24}\ell^3 \right] = \frac{m\omega^2}{2AY\ell} \frac{5}{24}\ell^3$$

$$\therefore \text{Ratio of strain} = \frac{\frac{m\omega^2}{2AY\ell} \times \frac{11\ell^3}{24}}{\frac{5m\omega^2}{2AY\ell} \times \frac{\ell}{24}} = \frac{11}{5}$$

$$= 2.20$$

2. **Ans (7.00)**

Since level of liquid in the cylinder remains same thus height remain same

$$\therefore \frac{V_{\text{new}}}{A_{\text{new}}} = \frac{V_{\text{old}}}{A_{\text{old}}}$$

$$\Rightarrow \frac{V_{\text{old}}(1 + \gamma_{\text{liq}}\Delta T)}{A_{\text{old}}(1 + 2\alpha\Delta T)} = \frac{V_{\text{old}}}{A_{\text{old}}}$$

$$\gamma_{\text{liq}} = 2\alpha = 2 \times 5 \times 10^{-7} = 10 \times 10^{-7} = 10^{-6}$$

$$\gamma + \beta = 7$$

3. Ans (5.00)

$$\text{Force exerted by the laser} = \frac{0.5 \times 10^{-3}}{c}$$

∴ Acceleration of the space craft

$$= \frac{F}{m} = \frac{0.5 \times 10^{-3}}{3 \times 10^8 \times 10^3} \text{ m/s}^2$$

∴ Time taken for velocity to become

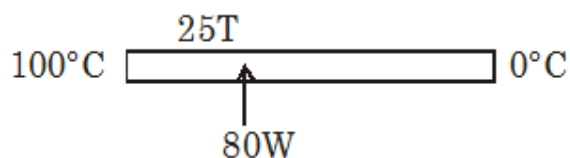
$$1 \text{ km/s} = \frac{10^{-3} \times 3 \times 10^{11}}{0.5 \times 10^{-3}} \text{ sec}$$

$$= 6 \times 10^{17}$$

$$\therefore \alpha = 6 \text{ \& } \beta = 17$$

$$\therefore (\beta - 2\alpha) = 5$$

4. Ans (105.00)



$$80 = \frac{kA}{0.25}(T - 100) + \frac{kA}{0.75}(T - 0)$$

$$80 = \frac{100}{4 \times 50} = T - 100 + \frac{T}{3}$$

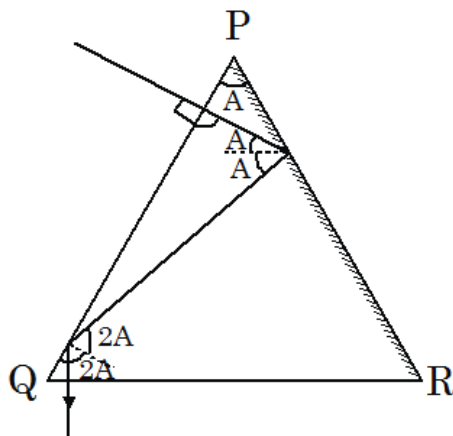
$$140 = \frac{4T}{3} \Rightarrow T = \frac{420}{4} = 105^\circ\text{C}$$

5. Ans (8.00)

One can then find V_3 (and, for that matter, $V_{\max}$) by using this, and get

$$\frac{V_3}{V_2} = \left(\frac{P_2}{P_3} \right)^{1/\gamma} = (32)^{3/5} = 8$$

6. Ans (9.00)



$$\frac{\pi}{2} - \frac{A}{2} + \frac{\pi}{2} - 2A = \frac{\pi}{2}$$

$$\frac{5A}{2} = \frac{\pi}{2}$$

$$A = \frac{\pi}{5} = 36^\circ$$

PART-2 : CHEMISTRY

SECTION-I (I)

1. Ans (A)

$$C_{v,m} = C_{p,m} - R \Rightarrow (13.314 - 8.314) + 0.04 T \Rightarrow 5 + 0.04 T$$

$$\Delta U = n.C_{v,m}.\Delta T$$

$$\Delta U = \left(\int_{T_1}^{T_2} (5 + 0.04T) \right) \times 2$$

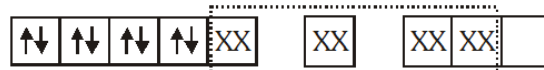
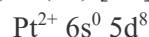
$$\Delta U = \left(5(\Delta T) + 0.4 \left(\frac{T_2^2}{2} - \frac{T_1^2}{2} \right) \right) \times 2$$

$$\Delta U = \left(5 \times 100 + \frac{0.04}{2} (160000 - 90000) \right) \times 2 = 3800 \text{ J}$$

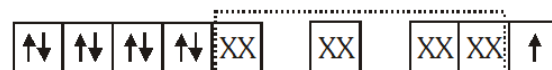
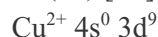
PART-2 : CHEMISTRY

SECTION-I (II)

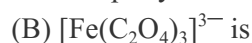
7. Ans (A,B,C)



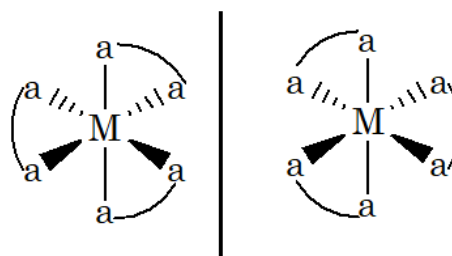
It has dsp^2 hybridisation, square planar shape.



It has dsp^2 hybridisation, square planar shape.



$M(a-a)_3$ type complex.



The mirror images are non superimposable.
Hence, optically active.

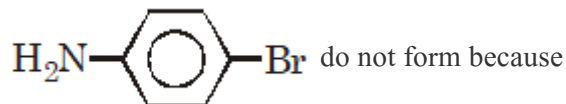
(C) Synergic bonding in metal carbonyl $\uparrow$,
CO bond strength decreases.

(D) $[\text{NiCl}_2(\text{PPh}_3)_2]$ is tetrahedral due to steric reasons, so No geometrical isomers.

8. **Ans (A,B,C,D)**

CO, C₂H₄, C₆H₆ & C₅H₅⁻ show synergic bonding.

10. **Ans (D)**

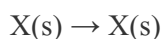


polar H on NH₂ so intramolecular reaction.

PART-2 : CHEMISTRY

SECTION-II

2. **Ans (1.00)**



$$\Delta G_2 - \Delta G_1 = \Delta V(P_2 - P_1)$$

$$0 - \Delta G^0 = \Delta V(6 - 1) \times 10^5$$

$$\Delta V = \left(\frac{20}{2.5} - \frac{20}{2} \right) = -2 \times 10^{-6} \text{ m}^3$$

3. **Ans (8.00)**

d⁰/d¹⁰ ⇒ Colourless.

PART-3 : MATHEMATICS

SECTION-I (I)

1. **Ans (A)**

$$A = \begin{bmatrix} 0 & \alpha \\ 0 & 0 \end{bmatrix}, A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(A+I)^{50} = I + 50A$$

$$\text{so } \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2. **Ans (B)**

Let vertices A₁, A₂, ... A₇ are 7th roots of unity.

Let A₁(1), A₂(α), A₃(α²), A₄(α³)

$$A_5 = \alpha^4 = \overline{\alpha^3}, A_6 = \alpha^5 = \overline{\alpha^2}, A_7 = \alpha^6 = \overline{\alpha}$$

$$(1 - \alpha)(1 - \alpha^2)(1 - \alpha^3)(1 - \alpha^4)(1 - \alpha^5)(1 - \alpha^6) = 7$$

$$\Rightarrow (|1 - \alpha||1 - \alpha^2||1 - \alpha^3|)^2 = 7$$

$$p_1 = (A_1A_2) (A_1A_3) (A_1A_4) (A_1A_5) (A_1A_6) (A_1A_7)$$

$$= |1 - \alpha||1 - \alpha^2||1 - \alpha^3||1 - \overline{\alpha^3}||1 - \overline{\alpha^2}||1 - \overline{\alpha}|$$

$$= |1 - \alpha||1 - \alpha^2||1 - \alpha^3||1 - \alpha^3||1 - \alpha^2||1 - \alpha|$$

$$= (|1 - \alpha||1 - \alpha^2||1 - \alpha^3|)^2 = (\sqrt{7})^2$$

Similarly

$$p_2 = |1 - \alpha||1 - \alpha^2||1 - \alpha^3||1 - \alpha^3||1 - \alpha^2|$$

$$= \sqrt{7} (|1 - \alpha^3||1 - \alpha^2|)$$

$$p_3 = \sqrt{7} (|1 - \alpha^3|)$$

$$p_4 = \sqrt{7}, p_5 = |1 - \alpha||1 - \alpha^2| \text{ and } p_6 = |1 - \alpha|$$

$$\Rightarrow p_1 \cdot p_2 \cdot p_3 \cdot p_4 \cdot p_5 \cdot p_6 = (\sqrt{7})^7$$

3. **Ans (C)**

$$\left| \frac{z - 2019}{z + 2019} \right| = 1 \Rightarrow z \text{ is purely imaginary} \Rightarrow z$$

lies on y-axis

Let A = (3, -4) and B = (5, 20) and P be any point on y-axis representing z.

|z - 3 + 4i| + |z - 5 - 20i| is minimum ⇒ PA + PB is minimum

Let A' = (-3, -4) = reflection of A in y-axis

join A'B. Its equation is y = 3x + 5

∴ It intersects y-axis at P(0, 5)

$$z = 5i \Rightarrow z^2 = -25$$

$$\Rightarrow 9z^2 = -225$$

$$\Rightarrow 9z^2 + 30 = -195$$

4. **Ans (A)**

$$\text{Let } x = x_1 + 1 \Rightarrow x_1 \geq 1. y = y_1 + 2 \Rightarrow y_1 \geq 1,$$

$$z = z_1 + 3 \Rightarrow z_1 \geq 1$$

$$\text{therefore the equation } x + y + z + w = 15 \Rightarrow$$

$$x_1 + y_1 + z_1 = 9 - w$$

Hence the number of positive integral

$$\text{solutions} = \sum_{w=1}^3 {}^{8-w}C_2$$

$$= {}^7C_2 + {}^6C_2 + {}^5C_2 = {}^8C_3 - {}^5C_3$$

$$= 56 - 10 = 46$$

PART-3 : MATHEMATICS

SECTION-I (II)

5. **Ans (A,C,D)**

$$a = \frac{7!}{2!} \times \frac{5!}{2!3!} = 25200$$

$$b = \frac{6!}{2!} \times \frac{5!}{2!3!} = 3600$$

when

$$\underbrace{\frac{9!}{2!3!}}_{\text{AA at end position}} + \underbrace{\frac{9!}{2!2!}}_{\text{EE at end position}} + \underbrace{\frac{9!}{2!2!} \cdot 2!}_{\text{A \& E at end position}} = 302400$$

6. **Ans (A,D)**

$$AB = A \Rightarrow ABA = A^2 \Rightarrow A^2 = A$$

Similarly $B^2 = B$

A and B are idempotent

$$\text{Also } (A - B)^2 = (A - B)(A - B)$$

$$= A^2 - AB - BA + B^2 = A - A - B + B = 0$$

(A - B) is nilpotent.

Correct options are (A) and (D).

7. **Ans (A,B,C)**

$$x^2 + y^2 = \frac{4}{(1 + a \sin t)^2}$$

$$\text{Also } ay = \frac{2a \sin t}{1 + a \sin t} = 2 - \frac{2}{1 + a \sin t}$$

$$\Rightarrow 2 - ay = \frac{2}{1 + a \sin t}$$

Squaring

$$(2 - ay)^2 = \frac{4}{(1 + a \sin t)^2} = x^2 + y^2$$

$\therefore$ Curve is

$$x^2 + (1 - a^2)y^2 + 4ay = 4$$

an ellipse if $a^2 - 1 < 0$

a parabola if $a^2 - 1 = 0$

a hyperbola if $a^2 - 1 > 0$

8. **Ans (A,B,C)**

$$|z_1|^2 = zz_1 \Rightarrow z_1 \bar{z}_1 = 1$$

$$\Rightarrow |z_1 + z_2 + z_3| = |\overline{z_1 + z_2 + z_3}| = |\bar{z}_1 + \bar{z}_2 + \bar{z}_3|$$

$$= \left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right| = |z_1 z_2 + z_2 z_3 + z_3 z_1|$$

$$\left(\frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_1)}{z_1 z_2 z_3} \right)$$

$$= \frac{(\bar{z}_1 + \bar{z}_2)(\bar{z}_2 + \bar{z}_3)(\bar{z}_3 + \bar{z}_1)}{\bar{z}_1 \bar{z}_2 \bar{z}_3}$$

$$= \frac{\left(\frac{1}{z_1} + \frac{1}{z_2}\right)\left(\frac{1}{z_2} + \frac{1}{z_3}\right)\left(\frac{1}{z_3} + \frac{1}{z_1}\right)}{\frac{1}{z_1 z_2 z_3}}$$

$$= \frac{(z_2 + z_1)(z_2 + z_3)(z_3 + z_1)}{z_1 z_2 z_3}$$

Since $\bar{z} = z \Rightarrow z$ is purely real and thus

$$\text{Im}(z) = 0$$

In option D clearly an isosceles right angled triangle happens

9. **Ans (B,D)**

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

total symmetric matrices = $3 \times 3 \times 3 = 27$

For singular matrices $|A| = 0$

$$\text{i.e. } ad - bc = 0$$

$$ad = bc$$

when $ad = bc = 1$ then no. of matrices = 1

when $ad = bc = 4$ then no. of matrices = 1

when $ad = bc = 9$ then no. of matrices = 1

when $ad = bc = 2$ then no. of matrices = 4

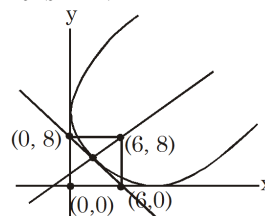
when $ad = bc = 3$ then no. of matrices = 4

when $ad = bc = 6$ then no. of matrices = 4

total number of singular matrices = 15

10. **Ans (A,B,C)**

A perpendicular dropped from focus to any tangent meet the tangent at the vertex circle is $(x - 6)x + y(y - 8) = 0 \Rightarrow x^2 - 6x + y^2 - 8y = 0$



$$\text{Equation of directrix is } y = \frac{-4}{3}x$$

$\therefore$ distance of (6, 8) from the directrix is

$$2a = \frac{48}{5}$$

$$\Rightarrow a = \frac{24}{5}$$

11. **Ans (A,C)**

$$(x - 2y - 7)(x + 2y + 5) = 4$$

$$\Rightarrow \frac{(x - 1)^2}{4} - \frac{(y + 3)^2}{1} = 1$$

Hyperbola :

$$\text{Auxiliary circle } (x - 1)^2 + (y + 3)^2 = 4$$

$$\Rightarrow x^2 + y^2 - 2x + 6y + 6 = 0$$

$$\text{Director circle } (x - 1)^2 + (y + 3)^2 = 4 - 1$$

$$x^2 + y^2 - 2x + 6y + 7 = 0$$

12. **Ans (A,B,D)**

n^{th} draw from n^{th} box results in a red.

All previous draws must result in white.

So

$$P(n) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{n-1}{n} \cdot \frac{1}{n+1} = \frac{1}{n(n+1)} < \frac{1}{2010}$$

$$\Rightarrow n(n+1) > 2010$$

$$\Rightarrow n \geq 45$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (0.60)**

solve $y^2 = 2x$ and $\frac{x^2}{9} + \frac{y^2}{4} = 1$ we get

$$P = \left(\frac{3}{2}, \sqrt{3}\right) \quad Q = \left(\frac{3}{2}, -\sqrt{3}\right)$$

tangent at P & Q to ellipse is

$$\frac{x}{6} \pm \frac{y}{4}(\sqrt{3}) = 1 \text{ which intersect at } R(6,0)$$

tangent at P & Q to parabola $y^2 = 2x$ will be

$$y(\pm 2\sqrt{3}) = \left(x + \frac{3}{2}\right) \text{ which intersect at}$$

$$\left(-\frac{3}{2}, 0\right) = S$$

$$A = \frac{1}{2} \left(6 + \frac{3}{2}\right) \times 2\sqrt{3} = \frac{15\sqrt{3}}{2}$$

2. **Ans (0.25)**

$$(y - 2)^2 = 4 \left(x - \frac{(11 - 3 \sin \theta - 4 \cos \theta)}{4} \right)$$

$$(y - 2)^2 = 4(x - \lambda) \quad \left(\lambda \in \left[\frac{3}{2}, 4 \right] \right)$$

$\lambda = 4$ due to given constraint.

Equation of tangent from origin is

$$y - 2 = m(x - 4) + \frac{1}{m}, \text{ this tangent passes}$$

through (0, 0)

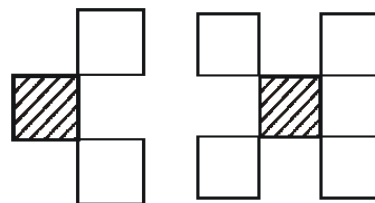
$$\Rightarrow m = \frac{1 \pm \sqrt{5}}{4}$$

$$\Rightarrow \left| \frac{1}{m_1 m_2} \right| = 4$$

3. **Ans (8.68)**

If 3 squares have 2 corners in common and no side in common then exactly one of the 3 squares will have both common corners as its own corners. Lets call such square a determining square.

A corner square can never be a determining square. For a square at the side there is exactly one configuration makes it a determining square. For a square not at the side there are ${}^4C_2 = 6$ configuration that makes it a determining square.



$$\text{Total} = 4 \times 0 + 24 \times 1 + 36 \times 6 = 240$$

$$\therefore \text{Probability} = \frac{240}{64C_3} = \frac{5}{868}$$

4. Ans (0.75)

$$\frac{x^2}{16} + \frac{y^2}{1} = 1$$

$$a^2 = 16, b^2 = 1$$

$$a = 4, b = 1$$

let $A(x_1, y_1)$ and $B(x_2, y_2)$ then P, Q will be

$$P\left(x_1, \frac{y_1}{4}\right) \text{ and } Q\left(x_2, \frac{y_2}{4}\right)$$

$$\text{mid point of AB will be } \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

mid point of PQ will be

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{8}\right) = (h, k)$$

$$\Rightarrow \text{mid point of AB will be } (h, 4k)$$

let point $S(t, t+8)$ be a point on $y = x + 8$

AB is chord of contact of $x^2 + y^2 = 16$

$$\Rightarrow xt + y(t+8) = 16 \quad \dots(i)$$

AB as chord whose mid point is $(h, 4k)$

$$xh + y(4k) = h^2 + 16k^2 \quad \dots(ii)$$

$$\frac{t}{h} = \frac{t+8}{4k} = \frac{16}{h^2 + 16k^2}$$

$$\Rightarrow t = \frac{16h}{h^2 + 16k^2} \text{ and } t+8 = \frac{64k}{h^2 + 16k^2}$$

$$\Rightarrow \frac{16h}{h^2 + 16k^2} + 8 = \frac{64k}{h^2 + 16k^2}$$

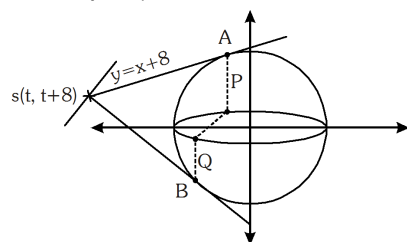
$$\Rightarrow 2h + (h^2 + 16k^2) = 8k$$

$$\Rightarrow x^2 + 16y^2 + 2x - 8y = 0$$

$$x^2 + \alpha y^2 + \beta x + \gamma y = 0$$

$$\Rightarrow \alpha = 16, \beta = 2, \gamma = -8$$

$$\Rightarrow \alpha - \beta + \gamma = 6$$



5. Ans (0.50)

We have $P(T) = p : P(H) = P(\bar{T}) = 1 - p$

$\therefore$ Required probability = $P(T \text{ or } HHT \text{ or } HHHHT \text{ or } \dots)$

$$= \frac{p}{1 - (1 - p)^2} = \frac{p}{2p - p^2} = \frac{2}{3}$$

$$\Rightarrow p = \frac{1}{2}$$

6. Ans (0.83)

$$\begin{vmatrix} 12 & b & c \\ a & 24 & c \\ a & b & 36 \end{vmatrix} = 0$$

$$(12)(24)(36) - 12bc - 36ba + 2abc - 24ac = 0$$

$\dots(i)$

$$\frac{1}{a-12} + \frac{2}{b-24} + \frac{3}{c-36} = \frac{(b-24)(c-36) + 2(a-12)(c-36) + 3(a-12)(b-24)}{(a-12)(b-24)(c-36)}$$

on solve

$$= -\frac{1}{6}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B	A,B,D	A,C	A,B	A,C	A,B,C	B,C	A,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	B	A	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	300.00	2.00	5.00	2.00	4.37 to 4.38	101.00 to 102.51	2.00	7.00

PART-2 : CHEMISTRY

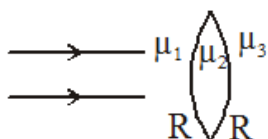
SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,C,D	A,B,C,D	A,B,D	A,B,D	B,C,D	B,D	A,B,D	A,B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	B	D	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	41.38	600.00	14.00	7.00	3.86 to 3.88	6.00	7.00	5.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	C,D	B,D	A,B	A,C,D	A,B	A,B,C	A,B,C,D	A,C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	B	C	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	1.80	1.75	1.65	1.71	2.25	3.60	2.50	0.25

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A,B)



Let us consider parallel rays,

 For refraction through 1st surface

$$\frac{\mu_2}{v_1} - \frac{\mu_1}{\infty} = \frac{\mu_2 - \mu_1}{R} \dots (1)$$

 For refraction through 2nd surface,

$$\frac{\mu_3}{v} - \frac{\mu_2}{v_1} = \frac{\mu_3 - \mu_2}{-R} \dots (2)$$

$$(1) + (2) \Rightarrow \frac{\mu_3}{v} = \frac{2\mu_2 - \mu_1 - \mu_3}{R}$$

 For lens to be diverging, $v \rightarrow -ve$

$$\Rightarrow 2\mu_2 - \mu_1 - \mu_3 < 0 \Rightarrow \mu_2 < \frac{\mu_1 + \mu_3}{2}$$

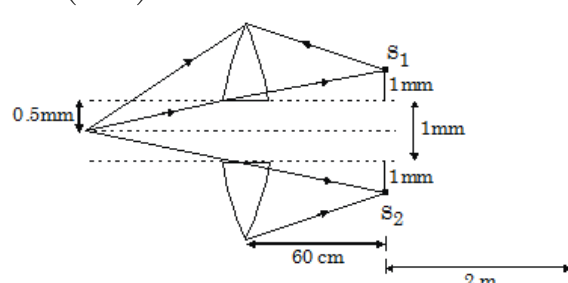
2. Ans (A,B,D)

For image

$$\frac{1}{v} + \frac{1}{30} = \frac{1}{20}$$

$$v = 60$$

$$m = \left(\frac{60}{-30} \right) = -2$$



$$\text{Fringe width} = \beta = \frac{D\lambda}{d} = \frac{2 \times 500 \times 10^{-9}}{3 \times 10^{-3}} \text{ m}$$

$$= \frac{1}{3} \text{ mm}$$

$$I = 2I_0 \text{ at } y = \frac{\beta}{4} = \frac{1}{12} \text{ mm}$$

$$\text{Distance of third minima is at } \frac{5\beta}{2} = \frac{5}{6} \text{ mm}$$

3. **Ans (A,C)**

$$\Delta x = \frac{\lambda}{2} + (2t)\mu_2$$

for constructive interference $\Delta x = n\lambda$

$$\text{for destructive interference } \Delta x = \left(n + \frac{1}{2}\right)\lambda$$

4. **Ans (A,B)**

$$\frac{\mu}{2R} - \frac{1}{-x} = \frac{\mu - 1}{R}$$

$$X = \frac{2R}{\mu - 2}$$

5. **Ans (A,C)**

$$K_{\max} = \frac{hc}{\lambda} - \phi; \quad I_{\text{ev}} = \frac{hc}{400} - 1.9$$

$$\text{So } hc = 2.9 \times 400 \text{ eV nm}$$

$$\text{Next } K_{\max} = \frac{hc}{500} - 1.9 = 0.42 \text{ eV}$$

$$\text{For longest } \lambda = \frac{hc}{\lambda_{\max}} = 1.9$$

6. **Ans (A,B,C)**

$$m = 100 \times 0.9 = 90 \text{ gm}$$

$$\frac{\Delta H}{\Delta t} = \frac{90 \times 80}{5 \times 60} = 24 \text{ cal/sec}$$

$$v_1 = \frac{90}{1} = 90 \text{ cc}$$

$$24 = ms \frac{dT}{dt}$$

$$24 = 90 \times 1 \times \frac{dT}{dt}$$

$$\frac{dT}{dt} = \frac{8}{30} \text{ }^\circ\text{C/sec.}$$

$$\frac{dV}{dt} = \frac{V}{3} r \frac{dT}{dt}$$

$$\tan \theta = 90 \times 10^{-3} \times \frac{8}{30} = 24 \times 10^{-3} \text{ cc/sec}$$

$$= 24 \times 10^{-3} \times 60 \text{ min.} = 1.44 \text{ cc/min.}$$

7. **Ans (B,C)**

$$\frac{dH}{dt} = \frac{kA}{L} (T_2 - T_1)$$

$$\frac{dH}{dt} = \frac{1 \times A}{1} (100 - T) = \frac{2 \times A}{1.5} (T - 0)$$

$$100 - T = \frac{4T}{3}$$

$$100 = \frac{7T}{3} \Rightarrow T = \frac{300}{7} \text{ }^\circ\text{C}$$

$$\text{for D } \frac{2 \times A}{1.25} (100 - T') = \frac{1 \times A}{0.5} (T' - 0)$$

$$\Rightarrow \frac{800}{5} - \frac{8T'}{5} = 2T'$$

$$160 = 3.6 T'$$

$$T' = \frac{1600}{36}$$

$$\frac{dH_1}{dt} = \frac{4}{3} A \times \frac{300}{7} = \frac{400A}{7}$$

$$\frac{dH_2}{dt} = \frac{800A}{21}$$

8. **Ans (A,C)**

$$dQ = dW + dU = \frac{3}{2} dW = 3dU$$

$$ncdT = 3n C_v dT$$

$$\Rightarrow C_v = \frac{5R}{2} = \frac{fR}{2}$$

Degree of freedom = 5 Diatomic

PART-1 : PHYSICS

SECTION-II

2. **Ans (2.00)**

$$E = \frac{n^2 h^2}{8mL^2}$$

$$\Delta E = \frac{(4-1)h^2}{8mL^2} = \frac{3h^2}{8mL^2}$$

$$\phi = \frac{h^2}{8mL^2}$$

$$(KE)_{\max} = \Delta E - \phi = 2 \frac{h^2}{8mL^2}$$

$$= 2 \times \text{Energy of electron at } n=1$$

$$\Rightarrow x = 2$$

3. **Ans (5.00)**

$$A = \frac{A_0}{2^n} \Rightarrow 2^n = \frac{6800}{425}; n = \text{No. of half life}$$

$$2^n = 16 = 2^4$$

$$\Rightarrow n = 4$$

$$\Rightarrow 4 \times t_{1/2} = 10$$

$$\Rightarrow t_{1/2} = 2.5 \text{ h}$$

4. **Ans (2.00)**

Initially let no. of molecules of A = N_A and that of B = N_B .

$$\frac{\lambda N_A}{\lambda N_A + 2\lambda N_B} = \frac{25}{100}$$

$$\therefore 100\lambda N_A = 25\lambda N_A + 50\lambda N_B \Rightarrow N_A = \frac{2}{3}N_B$$

$$\text{Now } N_A' = N_A e^{-\lambda t} = \frac{2}{3}N_B e^{-\lambda t}$$

$$N_B' = N_B e^{-2\lambda t}$$

$$\therefore \frac{\frac{2}{3}\lambda N_B e^{-\lambda t}}{\frac{2}{3}\lambda N_B e^{-\lambda t} + 2\lambda N_B e^{-2\lambda t}} = \frac{75}{100} = \frac{3}{4}$$

$$\Rightarrow \frac{1}{1 + 3e^{-\lambda t}} = \frac{3}{4}$$

$$\Rightarrow 1 = 9e^{-\lambda t} \Rightarrow e^{\lambda t} = 9$$

$$\text{So, } t = \frac{\ln(9)}{\lambda} \text{ hr} = \frac{2 \ln(3)}{\ln(3)} \text{ hr} = 2 \text{ hr}$$

5. **Ans (4.37 to 4.38)**

$$Q = (M_{Ne} - M_{Na})c^2$$

$$= 0.0047 \times 931.5 \text{ MeV} = 4.378 \text{ MeV}$$

6. **Ans (101.00 to 102.51)**

$$\lambda_e = \sqrt{\frac{150}{V}} \times 10^{-10} \text{ m}$$

$$\lambda = \frac{1240}{V} \times 10^{-9} \text{ m}$$

$$\sqrt{\frac{150}{V}} \times 10^{-10} = \frac{12400}{V} \times 10^{-11}$$

$$V = 102.50 \text{ to } 102.51 \text{ volts}$$

8. **Ans (7.00)**

$$-\frac{1}{A} \frac{dQ}{dt} = \sigma(T^4 - 0^4)$$

$$\Rightarrow -mc \frac{dT}{dt} = \sigma A T^4$$

$$\Rightarrow T^{-4} dT = -\frac{\sigma A}{mc} dt$$

$$\Rightarrow \int_{200K}^{100K} T^{-4} dT = -\frac{\sigma A}{mc} \int_0^t dt$$

$$\Rightarrow \frac{T^{-3}}{-3} \Big|_{200}^{100} = -\frac{\sigma A}{mc} t$$

$$\Rightarrow \frac{1}{(100)^3} - \frac{1}{(200)^3} = \frac{\sigma(4\pi r^2)}{\frac{4}{3}\pi r^3 \rho c}$$

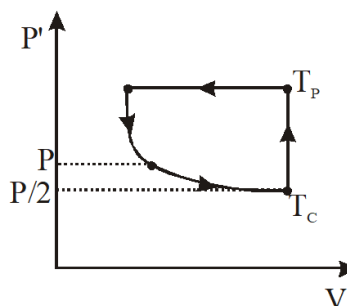
$$\Rightarrow t = \frac{7}{72} \frac{r \rho c}{\sigma} \times 10^{-6} \text{ s}$$

$$\Rightarrow t = \frac{7}{72} \frac{r \rho c}{\sigma} \mu \text{ s}$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. **Ans (A,B,C,D)**



$$\Delta S_{CD} = 4 \ln 16$$

$$= nC_v R \ln \frac{T_D}{T_C}$$

$$\frac{T_D}{T_C} = 16 = \frac{P}{P/2}$$

$$P' = 8P$$

$$\text{For AB } \frac{T_B}{T_A} = \left(\frac{P_A}{P_B} \right)^{\frac{1-\gamma}{\gamma}}$$

$$\frac{T_B}{300} = (8)^{-1/3} = \frac{1}{2}$$

$$T_B = 150$$

PART-2 : CHEMISTRY

SECTION-II

2. **Ans (600.00)**

Mole of X reacted = 0.75 mole

Heat liberated at constant pressure = 0.75×10 kJ = 7.5 kJ

mole of X remaining = $\frac{1}{4}$ mole

mole of Y formed = $\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$

$$\left(\frac{1}{4} C_{p,m(X)} + \frac{1}{2} C_{p,m(Y)} \right) (\Delta T) = q = 7500$$

$$\Delta T = \frac{7500}{\frac{40}{4} + \frac{30}{2}} = 300$$

Final temperature = $300 + 300 = 600 \text{ K}$

3. **Ans (14.00)**

$$\Delta G = nRT \ln \left(\frac{P_2}{P_1} \right) \Rightarrow 20 \ln 2 = 20 \times 0.7 =$$

14 Latm

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C,D)

$$|Z| = |Z - 1| \Rightarrow |Z|^2 = |Z - 1|^2 \Rightarrow Z + \bar{Z} = 1$$

$$\text{Similarly } |Z| = |Z + 1| \Rightarrow Z + \bar{Z} = 1$$

$$\text{Thus in both the cases } Z + \bar{Z} = 1$$

$$\text{Also } f(a) = 1 \text{ is false } \Rightarrow f(a) = 2 \text{ or } 3$$

$$f(b) \neq 1 \text{ is false } f(b) = 1$$

$$f(c) \neq 2 \text{ is true } f(c) = 1 \text{ or } 3$$

$$\text{But } f \text{ is one one}$$

$$\therefore f(b) = 1, f(c) = 3, f(a) = 2$$

2. Ans (B,D)

$$C \text{ is } \frac{x^2}{7} - \frac{y^2}{3} = 1 \quad (y \geq 0)$$

$$\therefore \text{Equation of L will be } y = x - 2 \text{ and}$$

$$A \left(\frac{7}{2}, \frac{3}{2} \right)$$

$$\text{Equation of circle}$$

$$\left(x - \frac{7}{2} \right)^2 + \left(y - \frac{3}{2} \right)^2 + \lambda(x - y - 2) = 0$$

$$\text{passing through } (0, 0)$$

$$\Rightarrow \lambda = \frac{29}{4}$$

3. Ans (A,B)

$$y = \frac{1}{3}x \pm \sqrt{\frac{1}{9}a^2 + b^2}$$

$$\Rightarrow -1 = -\frac{1}{3} \pm \sqrt{\frac{a^2}{9} + b^2}$$

$$\Rightarrow \frac{4}{9} = \frac{a^2}{9} + b^2$$

$$\Rightarrow \left(\frac{a^2}{9} + b^2 \right) \geq \left(\frac{a^2 b^2}{9} \right)^{1/2}$$

$$\frac{4}{9} \geq \left(\frac{ab}{3} \right) = \frac{2}{3} \geq ab = ab \leq \frac{2}{3}$$

$$\text{and use } 0 < e^2 < 1$$

$$a \in \left(\sqrt{\frac{2}{5}}, 2 \right)$$

4. Ans (A,C,D)

$$|\alpha - \beta|^2 = |\alpha|^2 + |\beta|^2$$

$$\Rightarrow \alpha\bar{\beta} + \beta\bar{\alpha} = 0$$

$$\text{and } \arg\left(\frac{\alpha}{\beta}\right) = \pm \frac{\pi}{2}$$

5. Ans (A,B)

$$\text{Delhi} \rightarrow \text{Chennai} \rightarrow \text{Dubai} \rightarrow \text{Sharjah} \rightarrow \text{Muscat}$$

$$P(\text{Luggage does not reach Sharjah})$$

$$1 - \left(1 - \frac{1}{20}\right) \left(1 - \frac{1}{10}\right) \left(1 - \frac{1}{5}\right) = 1 - \frac{19}{20} \cdot \frac{9}{10} \cdot \frac{4}{5}$$

$$= 1 - \frac{684}{1000} = \frac{316}{1000} = .316$$

$$P \left(\frac{\text{luggage was lost at Dubai}}{\text{luggage does not reach Muscat}} \right)$$

$$\frac{\left(1 - \frac{1}{20}\right) \cdot \left(1 - \frac{1}{10}\right) \cdot \frac{1}{5}}{1 - \left(1 - \frac{1}{20}\right) \left(1 - \frac{1}{10}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{10}\right)}$$

$$= \frac{855}{1922}$$

6. Ans (A,B,C)

$$\text{Since } 2A + 3BB' = I$$

$$\Rightarrow 2A = I - 3BB' \quad (\text{symmetric matrix})$$

$$\Rightarrow A \text{ is symmetric}$$

$$\Rightarrow B^{-1} = A' = A$$

$$\Rightarrow A \text{ \& B are inverse of each other}$$

$$\text{Also, } A^{-1} = B \text{ is symmetric}$$

$$\text{thus, } 2A + 3BB' = I \Rightarrow 2B^{-1} + 3B^2 = I$$

$$\Rightarrow 2I + 3B^3 = B \Rightarrow B - 3B^3 = 2I$$

$$(A) \text{tr}(A^{-1} - 3B^3 - BA + I)$$

$$= \text{tr}(B - 3B^3) = \text{tr}(2I) = 6$$

$$(B) \text{tr}(A^{-1} - 3B^3 + BA)$$

$$= \text{tr}(B - 3B^3 + I)$$

$$= \text{tr}(3I) = 9$$

$$(C) \text{Since, } 2A + 3BB' = I$$

$$\Rightarrow 2A + 3(A^{-1})^2 = I$$

$$\Rightarrow 2A^3 + 3I = A^2$$

$$\Rightarrow A^2 - 2A^3 = 3I$$

$$\text{So, } \det(A^2 - 2A^3) = \det(3I) = 27$$

$$(D) \det(A^{-1} - 3B^3 + BA) = \det(B - 3B^3 + I)$$

$$= \det(3I) = 27$$

7. Ans (A,B,C,D)

$$\text{Note that } A_\alpha A_\beta = A_{\alpha+\beta} \text{ \& } A_0 = I$$

$$\text{so, } A_\pi A_{\frac{3\pi}{2}} A_{\frac{\pi}{4}} \dots = A_S;$$

$$\text{Where } S = \pi + \frac{3\pi}{2} + \frac{5\pi}{4} + \dots$$

$$= \lim_{n \rightarrow \infty} \pi \left(1 + \frac{3}{2} + \frac{5}{4} + \frac{7}{8} + \frac{9}{16} + \dots \text{ upto infinite terms} \right)$$

$$= 6\pi$$

$$\text{so, } A_S = A_{6\pi} = I$$

$$\text{further } A_{(-\alpha)} = \text{adj}(A_\alpha)$$

8. **Ans (A,C,D)**

Equation of tangent to $y^2 = 4x$ at $(4, 4)$ is $x - 2y + 4 = 0$

Now, distance of the center of the variable circle from this tangent is equal to its distance from $(1, 0)$. Hence, locus of the center will be a parabola having focus at $(1, 0)$ & $x - 2y + 4 = 0$ as the directrix.

So, length of latus rectum

$$= 2 \times \frac{5}{\sqrt{5}} = 2\sqrt{5}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

9. **Ans (C)**

(I) Required probability = $\frac{1}{2}$

(II) Required probability
 $= \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 - \left(\frac{1}{2}\right)^7 = \frac{47}{2^7}$

(III) Required probability = $\frac{4}{2^7}$

(IV) There are five cases 0, 1, 2, 3, or 4 wins
 Required probability
 $= \frac{1 + 7 + 15 + 10 + 1}{2^7} = \frac{34}{2^7}$

10. **Ans (B)**

(I) Required probability = $\frac{1}{2}$

(II) Required probability
 $= \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 - \left(\frac{1}{2}\right)^7 = \frac{47}{2^7}$

(III) Required probability = $\frac{4}{2^7}$

(IV) There are five cases 0, 1, 2, 3, or 4 wins
 Required probability
 $= \frac{1 + 7 + 15 + 10 + 1}{2^7} = \frac{34}{2^7}$

11. **Ans (C)**

(I) ${}^9C_5 + {}^9C_4 + {}^9C_4 + {}^9C_3$
 $= {}^{10}C_5 + {}^{10}C_4 = {}^{11}C_5 = {}^{11}C_6$

(II) $= {}^{10}C_5 + {}^{10}C_4 = {}^{11}C_5 = {}^{11}C_6$

(III) ${}^9C_5 + {}^9C_5 = 2 \cdot {}^9C_5$
 $= {}^9C_5 + {}^9C_4 = {}^{10}C_5$

(IV) ${}^{10}C_5 = 2 \cdot {}^9C_5$

12. **Ans (B)**

(I) ${}^9C_5 + {}^9C_4 + {}^9C_4 + {}^9C_3$
 $= {}^{10}C_5 + {}^{10}C_4 = {}^{11}C_5 = {}^{11}C_6$

(II) $= {}^{10}C_5 + {}^{10}C_4 = {}^{11}C_5 = {}^{11}C_6$

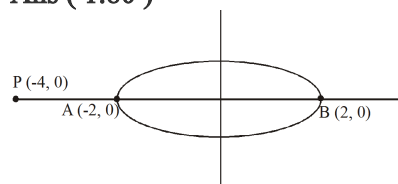
(III) ${}^9C_5 + {}^9C_5 = 2 \cdot {}^9C_5$
 $= {}^9C_5 + {}^9C_4 = {}^{10}C_5$

(IV) ${}^{10}C_5 = 2 \cdot {}^9C_5$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1.80)**



Given curve is $\frac{(x-2)^2}{4} + \frac{(y+1)^2}{1} = 1$

$$\Rightarrow \frac{X^2}{4} + \frac{Y^2}{1} = 1$$

Now $\frac{\max(X+4)^2 + Y^2}{\min(X+4)^2 + Y^2}$
 $= \frac{\text{square of max distance from P to any point on the ellipse}}{\text{square of min distance from P to any point on the ellipse}}$
 $= \frac{(PB)^2}{(PA)^2} = 9$

2. **Ans (1.75)**

There are four favourable combinations of three numbers

$$7 + 8 + 9, 6 + 8 + 9, 5 + 8 + 9, 6 + 7 + 9$$

Hence, $n(E) = 4 \times (3 \times 3! \times 6!)$ & $n(S) = 9!$

3. **Ans (1.65)**

$x < y < z \Rightarrow$ number of numbers $= {}^9C_3$

$x = y < z \Rightarrow$ number of numbers $= {}^9C_2$

$x < y = z \Rightarrow$ number of numbers $= {}^9C_2$

$x = y = z \Rightarrow$ number of numbers $= 9$

total numbers $= {}^9C_3 + {}^9C_2 + {}^9C_2 + 9 = 165$

4. **Ans (1.71)**

$$4^a + 4^b + 3$$

$$= (5-1)^a + (5-1)^b + 3$$

$$= 5 (\text{integer}) + (-1)^a + (-1)^b + 3$$

$\Rightarrow$ a and b both must be even

$$\Rightarrow \text{prob} = \frac{{}^{25}C_2}{{}^{50}C_2} = \frac{25 \times 24}{50 \times 49} = \frac{12}{49}$$

5. Ans (2.25)

Each term independently zero

$$\Rightarrow a = \frac{\pi}{4}, \quad b = 1, \quad c = 0$$

$$\Rightarrow D = \begin{vmatrix} 1 & 1/2 & 1 \\ 3 & 1 & 1 \\ 0 & 5 & 2 \end{vmatrix} = 9 \Rightarrow \frac{D}{3} = 3$$

6. Ans (3.60)

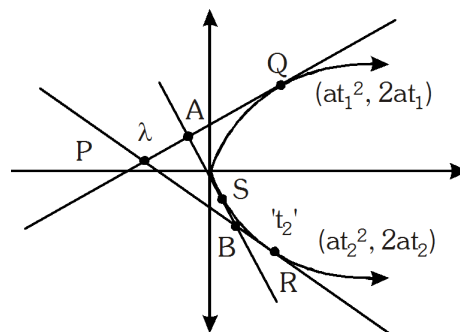
$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix} = \begin{vmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{vmatrix}$$
$$= - \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{vmatrix} = (1^3 + 2^3 + 3^3 - 3(1)(2)(3))$$
$$= 18$$

7. Ans (2.50)

$$\omega^{23} = 1$$

$$\begin{aligned} N &= \sum_{k=1}^{22} \frac{1}{1 + \omega^{8k} + \omega^{16k}} \\ &= \sum_{k=1}^{22} \left(\frac{1 - \omega^{8k}}{1 - \omega^{24k}} \right) \\ &= \sum_{k=1}^{22} \left(\frac{1 - \omega^{8k}}{1 - \omega^k} \right) \\ &= \sum_{k=1}^{22} [1 + \omega^k + \omega^{2k} + \dots + \omega^{7k}] \\ &= 22 + \sum_{k=1}^{22} (\omega^k) + \sum_{k=1}^{22} (\omega^{2k}) + \dots + \sum_{k=1}^{22} (\omega^{7k}) \\ &= 22 - 7 = 15 \end{aligned}$$

8. Ans (0.25)



$$P(a(t_1), t_2, a(t_1 + t_2))$$

$$S(at_3^2, 2at_3)$$

$$A(at_1t_3, a(t_1 + t_3))$$

$$B(at_2t_3, a(t_2 + t_3))$$

Let $PA : PQ = \lambda : 1$

Using section formula

$$\lambda = \frac{t_3 - t_2}{t_1 - t_2}$$

$$\Rightarrow \frac{PA}{PQ} = \frac{t_3 - t_2}{t_1 - t_2}$$

Similarly $\frac{PB}{PR} = \frac{t_1 - t_3}{t_1 - t_2}$

$$\Rightarrow \frac{PA}{PQ} + \frac{PB}{PR} = 1$$